Time Series Analysis of Tweets related to TV programs
－The relationship between posting tweets and watching TV－

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Abstract

Fundamental statistics of tweets mentioned about NHK TV programs during the broadcasting are investigated, and we found that the partial autocorrelation function (PACF) and anomaly in the tweet time series influence somehow the total tweet volume of each program. By applying vector autoregression (VAR) model to time series of tweets and audience rate, both directions of influence between them are observed. We postulate the dominant direction depends on the composition of the program as well as the program genre.

Keyword: Time series analysis, Vector Auto Regression model

1. Background of the research

Today, some people try to get the image of real word by analyzing social data, and another think about using voices in the net space to make a movement and change the real word. For TV viewing, these two topics are interested in; 1. “Does the tweet number reflect the action of watching TV?” 2. “Does the climax of the tweets introduce people to watching TV program?” As for the first point, it is known and we also confirmed that only small correlation exists between total tweet volume and average audience rate among the programs. As for the second, the long-established rating company Nielsen has released the statement[1], "A spike in TV ratings can increase the volume of Tweets, and, conversely, a spike in Tweets can increase tune-in".

2. Experiments and results

We first collected the tweets posted during the broadcasting about one documentary series which contains 120 programs, and investigated stability, PACF, and number of abnormal values appeared in each time series. We found that larger value of PACF is essential to get the large amount of total tweets, and frequent spiking also related somehow to the total tweet volumes. Second, being applied VAR model to the series of tweets per min. and audience rate per min. on 100 TV programs, which cover most of all TV program genre (drama, music, variety show, educational program and documentary), we confirmed both direction of influences between tweet and audience rate exist.

3. Discussions

The frequency of appearance, of those two direction in a program series, is different even in the same program genre. The tweet text suggests that the main role of the tweets seems different in each direction, and we presumed that the program composition determines either direction of influence will appear or not.

4. Future Work

Further study on our hypothesis from the viewpoint of natural language processing, such as finding the topics chatted during the broadcasting is planned; It may reveal the lifetime of the topics. We also interested in studying tweet transition after broadcasting.

References

[1] http://www.nielsen.com/us/en/insights/news/2013/the-follow-back--understanding-the-two-way-causal-influence-betw.html Aug. 6, 2013